

STELLWAGEN BUOY 44029, MA, USA

WMO: 997140

Lat: 42.523N

Lon: 70.566W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 06-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.8	-8.5	N/A	N/A	N/A	N/A	N/A	N/A	17.1	-3.5	16.1	-4.2	10.2	310	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	23.5	N/A	22.7	N/A	21.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	13.2	11.2	DB	-13.9	25.1	1.4	1.5	-14.9	26.2	-15.7	27.1	-16.5	28.0	-17.6	29.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-0.2	-0.2	2.1	6.6	11.3	16.1	19.7	19.7	17.4	12.6	7.8	3.9	(6)	
	DBStd	7.76	4.76	3.86	3.32	2.38	2.42	2.49	1.74	1.50	2.19	2.81	3.42	3.85	(7)	
	HDD10.0	1248	317	287	244	105	13	1	0	0	0	11	81	190	(8)	
	HDD18.3	3241	576	520	503	352	219	75	6	5	43	177	317	448	(9)	
	CDD10.0	1168	0	0	1	3	53	182	301	301	222	92	13	1	(10)	
	CDD18.3	119	0	0	0	0	1	7	49	48	15	0	0	0	(11)	
	CDH23.3	34	0	0	0	0	0	2	24	5	3	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.9	7.9	7.5	6.7	5.6	4.9	4.5	4.0	4.2	5.0	6.3	6.8	7.4	(14)	
(15) Precipitation	PrecAvg	1210	99	89	113	103	92	99	85	89	102	115	101	115	(15)	
	PrecMax	1668	166	216	421	257	427	286	215	202	223	338	202	186	(16)	
	PrecMin	861	28	41	18	11	16	13	31	25	41	5	29	37	(17)	
	PrecStd	194	31	38	82	55	77	64	49	49	52	82	41	38	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	8.2	11.7	14.2	19.4	23.0	25.4	24.0	23.5	19.3	15.4	12.3	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	8.7	6.9	8.6	11.9	16.9	21.1	23.9	23.0	22.3	17.9	13.6	11.0	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	7.6	5.7	7.1	10.6	15.7	20.0	22.9	22.4	21.3	16.9	12.8	10.2	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	6.4	4.9	5.9	9.7	14.6	19.1	22.1	21.8	20.3	16.1	12.0	9.3	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.2	4.3	3.3	3.3	3.0	3.2	3.0	2.8	3.4	3.7	4.7	(35)	
		MCDBR	5.5	5.1	4.8	4.6	4.9	4.1	4.0	3.5	3.4	3.3	3.6	5.1	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.304	0.324	0.367	0.403	0.452	0.470	0.434	0.378	0.337	0.313	0.298	(40)	
	taud		2.530	2.521	2.512	2.418	2.361	2.257	2.211	2.303	2.442	2.550	2.577	2.577	(41)	
	Ebn at Noon		855	907	925	902	873	828	807	826	854	855	830	825	(42)	
	Edh at Noon		70	82	94	111	121	135	140	123	100	80	67	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.68	2.56	3.58	4.65	5.23	5.63	5.94	5.34	4.26	2.84	1.88	1.39	(44)	
	RadStd		0.11	0.22	0.24	0.50	0.56	0.51	0.37	0.20	0.27	0.23	0.15	0.16	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	+42

Nomenclature: See separate page